Jiahui Hu

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Education

South China University of Technology (SCUT) Sep 2024 – Jun 2028 (Expected)

Undergraduate, **Computer Science & Finance** GPA: **3.7 / 4.00** Average: **87.79 / 100** Rank: **3/15**

Research Experience

Research Assistant Sep 2025 – Present
EPIC-Lab, Shanghai Jiao Tong University *Advisor: Prof. Linfeng Zhang*

Research Projects

TempoKV: Predictive KV Cache Quantization for Autoregressive Video Generation

- Identified an asymmetric predictive structure in autoregressive video KV caches: Keys are highly predictable from temporally aligned Key history, whereas Value states are better predicted from the current aligned Key together with recent Value history.
- Developed **TempoKV**, a predictive-residual quantization framework that predicts each KV cache from reconstructed context and quantizes only the remaining residual. Achieved SOTA quality-efficiency trade-offs on different backbones (Self-Forcing, Causal Forcing, LongCat-Video-13B)
- Implemented the method with custom CUDA and Triton kernels, reducing the resident KV cache from **51.8 GB to 7.6 GB** and peak GPU memory from 71.5 GB to **28.7 GB** on Causal Forcing; validated across three backbones at horizons up to **1401 frames**.

Publications

[1] **Jiahui Hu***, Weile Mo*, Peiliang Cai, Jiacheng Liu, Jinkui Ren, Xiantao Zhang, Xiaobing Tu, Linfeng Zhang. *"TempoKV: Predictive KV Cache Quantization for Autoregressive Video Generation."*
Under review, ICLR 2027. arXiv:2610.15324.

[2] Jinshan Liu*, Haoran Qin*, Xiaobing Tu, Jiacheng Liu, **Jiahui Hu**, *et al.* *"LinCa: Accelerating Diffusion Models via Learnable Decomposed Feature Caching."*
ECCV 2026. arXiv:2608.17973.

[3] Zexuan Yan, Jiarui Jin, Yue Ma, Shijian Wang, **Jiahui Hu**, Wenxiang Jiao, Yuan Lu, Linfeng Zhang. *"Glyph-Banana: Advancing Precise Text Rendering Through Agentic Workflows."*
Under review, AAAI 2027. arXiv:2603.12155.

Awards

- **Meritorious Winner (Top 7%)**, Mathematical Contest in Modeling (MCM/ICM), 2025.
- **First-class Scholarship (Top 10%)**, South China University of Technology, 2025.
- **Second Prize**, Chinese Mathematics Competitions, 2025.

Skills

Programming	Python, C++, LaTeX, Bash, CUDA
Systems	Tools: Linux, Windows, Git, Slurm
Language	English: CET-6 590